

**A****DATA PREPARATION***(outside CV, label-free)***Raw aCGH Data**

100 samples x 2,834 regions  
 Discrete CN: {-1, 0, +1, +2}

**Region Merging**

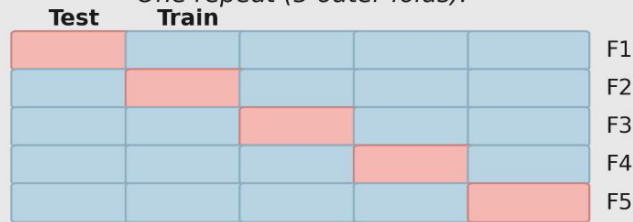
Pearson  $r > 0.8$  threshold  
 Label-free (no target leakage)

**Merged Data**

100 samples x 273 segments  
*Input to nested CV*

**B****REPEATED STRATIFIED 5-FOLD CV**

200 repeats (seeds 1001 - 1200)

*One repeat (5 outer folds):***Within each outer fold:****80% Train**

Fit hierarchical  
 classifier

**20% Test**

Evaluate:  
 BA (3-class)

**5-Fold Inner CV**

Tunes Stage 2  
*Scoring: balanced acc.*

**Primary metric:**  
 Stage 2 BA (BA2)

**200 repeats x 5 folds = 1,000 estimates***Report: mean +/- std***C****HIERARCHICAL CLASSIFIER****Test samples****STAGE 1**

HER2+ vs Rest (binary)

**BA=1.0**

KW top-5 + Random Forest  
 500 trees, balanced weights  
**FIXED across all pipelines**

**HER2+****HER2+****Rest****STAGE 2**

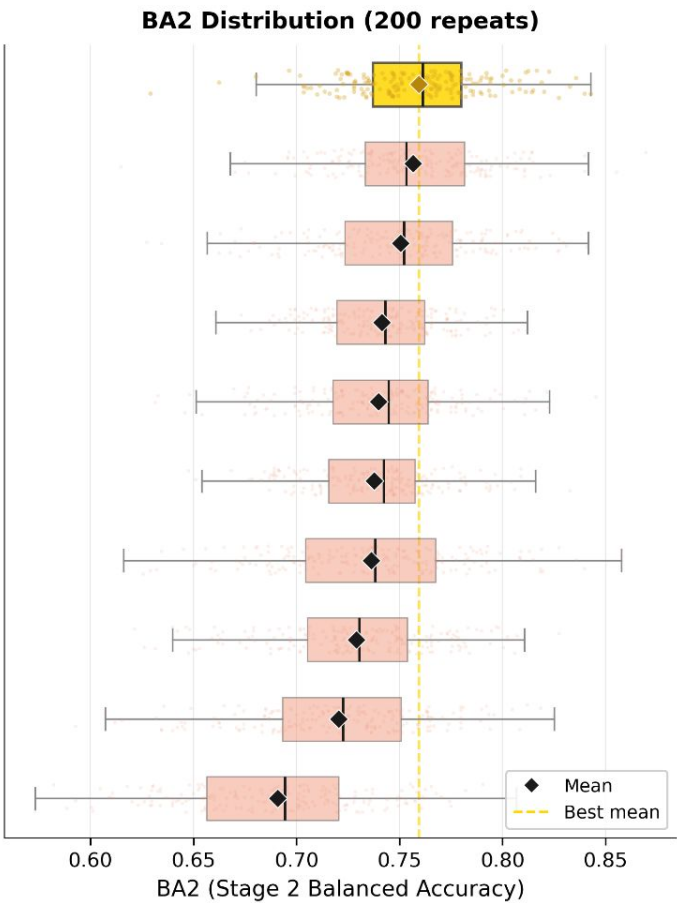
HR+ vs Triple Neg (binary)

Feature selection + Classifier  
**TUNED via inner CV**  
 4 pipeline variants compared

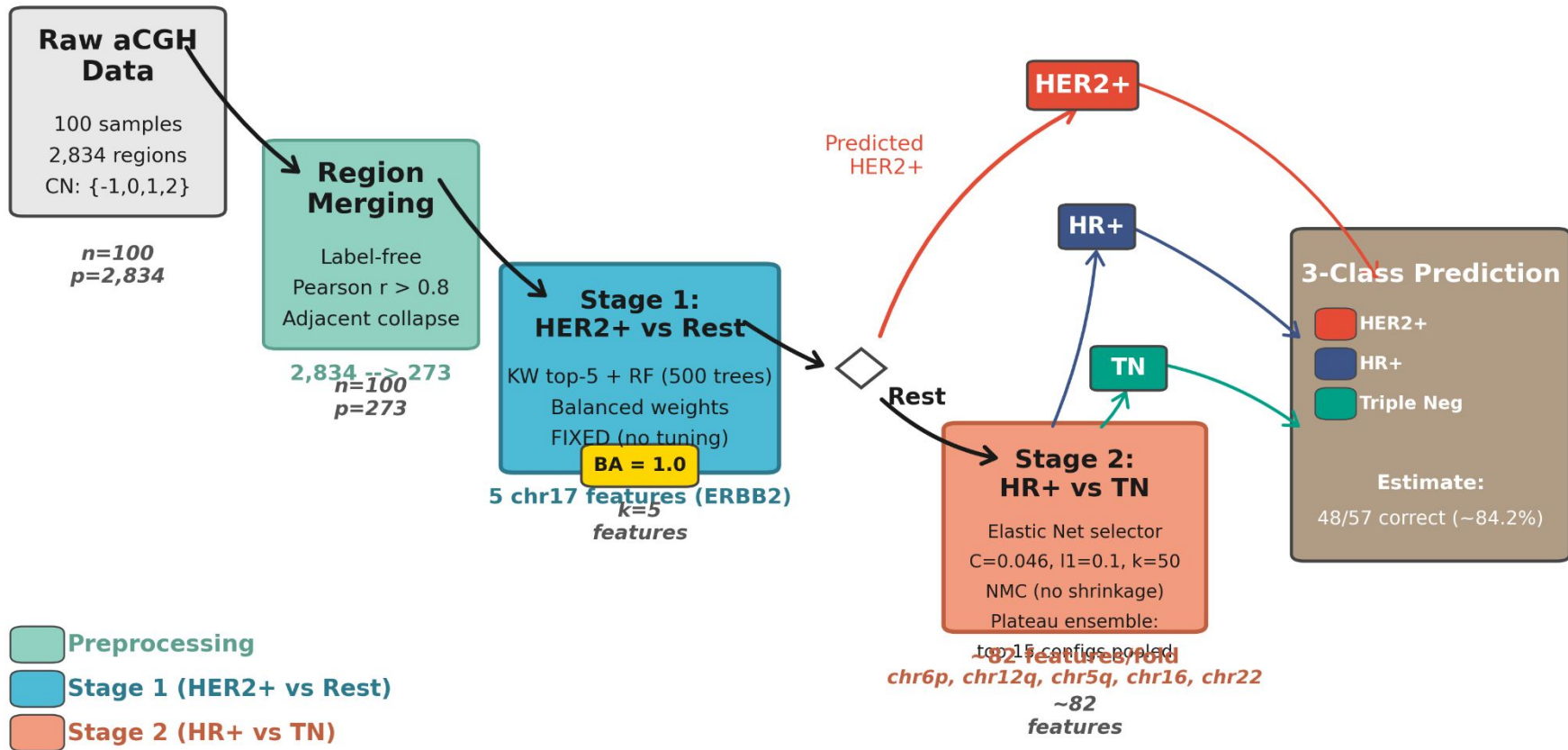
**HR+****Triple Neg****3-class prediction**

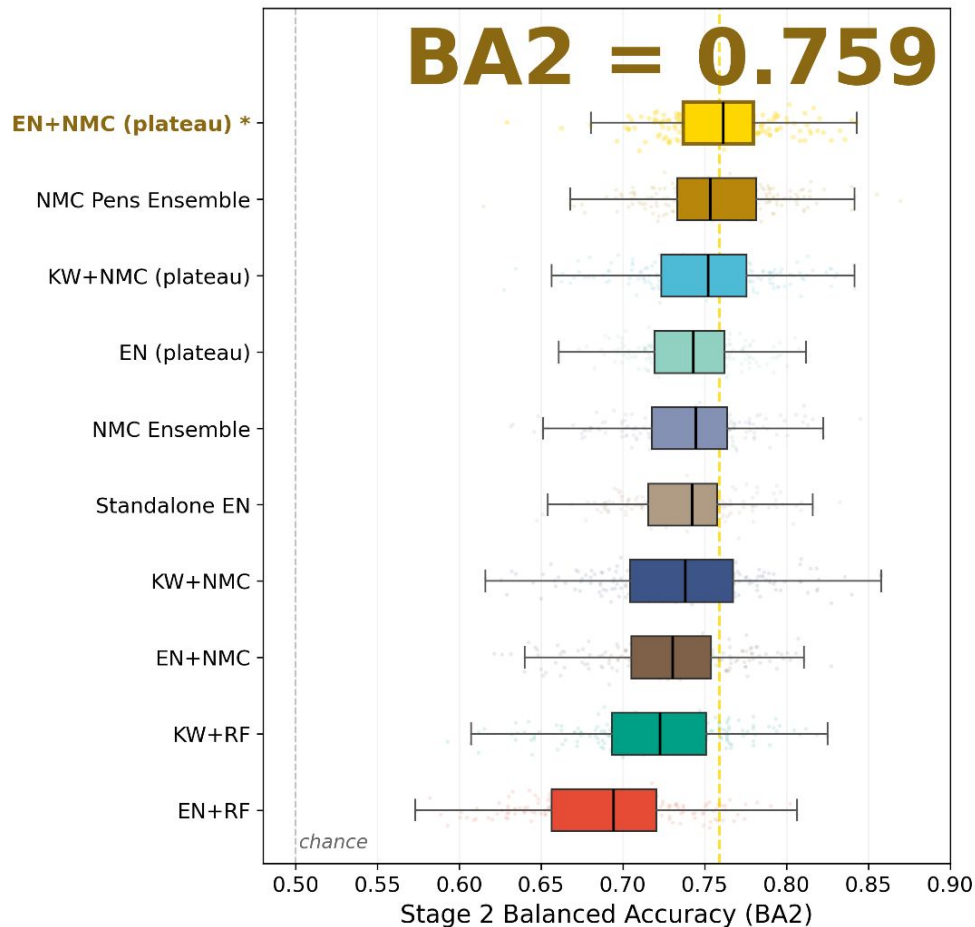
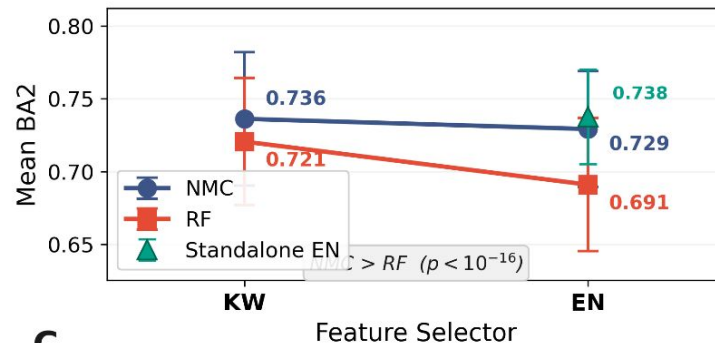
| #  | Pipeline              | BA2<br>(mean +/- std) | Combined BA<br>(mean +/- std) | AUROC | Acc.  | Feat. |
|----|-----------------------|-----------------------|-------------------------------|-------|-------|-------|
| 1  | EN + NMC (plat.) *    | 0.759 +/- 0.032       | 0.840 +/- 0.022               | 0.951 | 0.837 | 82.3  |
| 2  | NMC Pens Ens.         | 0.757 +/- 0.037       | 0.838 +/- 0.025               | 0.950 | 0.835 | -     |
| 3  | KW + NMC (plat.)      | 0.750 +/- 0.040       | 0.834 +/- 0.026               | 0.946 | 0.831 | 100.0 |
| 4  | Standalone EN (plat.) | 0.741 +/- 0.029       | 0.828 +/- 0.019               | 0.944 | 0.826 | -     |
| 5  | NMC Ensemble          | 0.740 +/- 0.037       | 0.827 +/- 0.025               | 0.941 | 0.823 | -     |
| 6  | Standalone EN         | 0.738 +/- 0.033       | 0.825 +/- 0.022               | 0.938 | 0.823 | -     |
| 7  | KW + NMC              | 0.736 +/- 0.046       | 0.824 +/- 0.031               | 0.938 | 0.821 | 41.6  |
| 8  | EN + NMC              | 0.729 +/- 0.040       | 0.819 +/- 0.026               | 0.935 | 0.816 | 32.5  |
| 9  | KW + RF               | 0.721 +/- 0.044       | 0.814 +/- 0.029               | 0.932 | 0.810 | 46.9  |
| 10 | EN + RF               | 0.691 +/- 0.046       | 0.794 +/- 0.031               | 0.919 | 0.791 | 32.4  |

\* Submitted method



BA2 = Stage 2 balanced accuracy (HR+ vs TN). Combined BA includes perfect Stage 1 (HER2+ vs rest). 200 repeated stratified 5-fold CV.



**A****B****C**

## Why EN+NMC (plateau)?

### 1. Simple classifier wins.

NMC > RF across all selectors ( $p < 10^{-16}$ ).  
Fewer parameters resist overfitting ( $n = 72$ ).

### 2. Plateau exposes true EN+NMC quality.

Base EN+NMC (0.729) < standalone EN (0.738): inner CV noisy at small  $n$ . Plateau pools top configs  $\rightarrow$  0.759.

### 3. EN+NMC(P) outperforms standalone EN(P).

Best model overall (0.759 vs 0.741): separating selection from classification yields higher BA2.

### 4. Aligns with Wessels et al. (2005).

Simple classifiers with univariate filtering match or exceed complex pipelines on small- $n$  discrete data.